## Jargon Buster

**High-pass filter:** A filtering circuit that only lets through frequencies higher than a specified frequency, blocking all lower frequencies. Acts like a check-dam in the data stream.
**Low-pass filter:** Passes frequencies lower than a specified frequency. Acts like a height barrier on roads before low overbridges.
**ATU:** ADSL Transceiver or Termination Unit, respectively, depending on whether it is installed with the telco, or the household end.
**Loading Coils:** An inductor placed on the local loop by the phone company. Coils are generally placed between 3,000 feet and 6,000 feet intervals to suppress exactly the signal that DSL modems need to transmit high speed data on—high frequency. The effect of a load coil is similar, from the perspective of the DSL equipment, to adding 20k feet to the line length.

No DSL equipment now works through load coils.
**Unterminated pairs:** a pair of twisted or untwisted cables not bridged or joined at the end that can cause disturbance and static on phone lines.
**Cross-talk:** Noise from one telephone line that is audible on another line. Caused when lines are too close, touching, not twisted, or by bad punch-downs or other wiring problems. This can cause noise bleed—through from the DSL line to other voice lines in the same location.

bridged taps (un-terminated pairs), and cross-coupled interference (cross talk between wires in the feeder cables that run to local exchanges). All this effectively restricts the available bandwidth in copper wire cables.

### Splitting the bottleneck, splaying the bandwidth

Telephone wires are bundled together in large cables, and 50 pairs to a cable is a typical configuration. Cables coming out of a central office (CO), though, may have hundreds or even thousands of pairs bundled together. An individual line from a CO to a subscriber is spliced together from many cable sections, as they fan out from the CO.

The normal copper wiring for POTS can support bandwidths of up to 1.1 MHz, but since voice signals are only transmitted in the lowest end of the spectrum, only the lowest 4 KHz is reserved for voice. DSL widens the bottleneck by utilising the frequencies above the 4 KHz range. The rest of the bandwidth is split into two bands, one high-speed downstream band, and another medium-speed duplex band.

Interestingly, the 'upstream' data rate, or data coming from the client computer to the server, is very low compared to the downstream. Typically, if the downstream is about 6 Mbps, the upstream would at most be 128 Kbps. This figure varies according to the service subscribed to.

There is but one important difficulty with twisted pair copper wire. It creates interference and cross talk, and the problem worsens with increasing distance and length of wire. It turns out in the end that attempts to send symmetric signals in many pairs within a single cable may severely limit the data rate and length of line attainable. Asymmetric DSL, in context, is ideally suited to serve high-speed data over short distances. The reason for this is that the data rate falls with distance from the CO. The table below gives a rough idea of the data rate ADSL media can transport over given distances.

As can be seen, the data transfer rate falls off with distance, which makes it difficult to deploy ADSL as a long-range broadband solution. For the same reason, it can be a feasible short-range solution for Internet over telephone wires.

### Challenge of the racing data

Transmitting data at tearaway rates as ADSL does, poses a host of technical challenges. ADSL modems divide the available bandwidth of a telephone line in one of two ways—frequency-division multiplexing (FDM) or echo cancellation *(see figures on page 99)*. This is known as Discrete Multitone Modulation (DMT). DMT is a modulation technique using FDM within the DSL band. Echo cancellation assigns the upstream band to overlap the downstream, and separates the two by means of local echo cancellation, a technique well known

| Data rate (Mbps) | Wire gauge (AWG) | Distance (feet) | Wire size (mm) | Distance (kilometers) |
|---|---|---|---|---|
| 1.5 or 2 | 24 | 18,000 | 0.5 | 5.5 |
| 1.5 or 2 | 26 | 15,000 | 0.4 | 4.6 |
| 6.1 | 24 | 12,000 | 0.5 | 3.7 |
| 6.1 | 26 | 9,000 | 0.4 | 2.7 |

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*Data rate drops with distance. This table shows how much, taking into considerating wire gauge as well.*